AIR UNIVERSITY ISLAMABAD

Kamra Campus • Department of Computer Science

LAB TASK 4: INTERACTIVE PRODUCT CUSTOMIZATION PLATFORM

Course: Full Stack Development (CS-301L) | Semester: BSCS VB | Instructor: M. Fahad Qasim

Student Name: Hamza Tahir

Registration ID: 225193

Discipline: BS Computer Science

Submission Date: October 2026

Technology Stack: HTML5, CSS3, JavaScript (ES6+), Lottie

Status: All 15 Tasks Completed & Verified

1. Executive Overview & Requirements Matrix

This lab assignment implements a modern, high-performance eCommerce product configurator for customized ceramic mugs. The application features rich interactive mouse interactions, real-time preview manipulation, HTML5 drag-and-drop, simulated payment and card validation, dynamic state unlocking, Lottie animations, and interactive SVG graphics.

No.	Requirement Specification	Technical Implementation Details	Status
1	Product Information & Copy Protection	Name, price, descriptive tags; JavaScript `copy` and `cut` event prevention with visual toast alert feedback.	Completed
2	Product Customization Controls	Real-time custom text engraving, 18-character limit counter, size variants (Small, Medium, Large), color glaze filters.	Completed
3	Interactive 3D Product Preview	CSS 3D perspective transforms with mouse coordinates calculation (`rotateX`, `rotateY`) and specular highlight gradient overlay.	Completed
4	Product Drag-and-Drop	Native HTML5 Drag and Drop API with dragover highlight states, drop sound/visual pulse, cart inventory update, and confirmation banners.	Completed
5	Video Advertisement	Integrated HTML5 `<video>` element with playback controls, styled responsive container, and 16:9 aspect ratio.	Completed
6	Animations & Visual Effects	Smooth CSS cubic-bezier transitions, pulsing status badges, steam SVG keyframe animations, and card hover elevations.	Completed

No.	Requirement Specification	Technical Implementation Details	Status
7	Lottie Animation	Integrated `lottie-web` vector celebration animation triggered upon successful order placement and checkout.	**Completed**
8 & 9	Premium Customization & Feature Lock	4 exclusive artisan glazes, graphic stamps, and cursive luxury typography styles locked with visual indicators and modal triggers.	**Completed**
10 & 11	Premium Payment & Dynamic Status	Simulated card gateway modal; dynamic DOM update from "Free User" to "Premium User" without page reload upon completion.	**Completed**
12	Form Validation & Submissions	Strict regex validation for cardholder name, 16-digit card number with grouping, expiry MM/YY format, and CVV checks.	**Completed**
13	Form & Application State Reset	"Reset All Customizations" restores all inputs, visual glaze filters, badges, preview states, and cart items to original defaults.	**Completed**
14	Bonus Interactive SVG Graphics	Interactive vector mug and VIP Authenticity Seal that dynamically change fill colors and rotate based on user interaction.	**Completed**
15	Responsive Layout Across Devices	CSS Grid and Flexbox media queries ensuring seamless responsiveness on desktop, tablet, and mobile screens.	**Completed**

2. Application Screenshots & Feature Demonstration

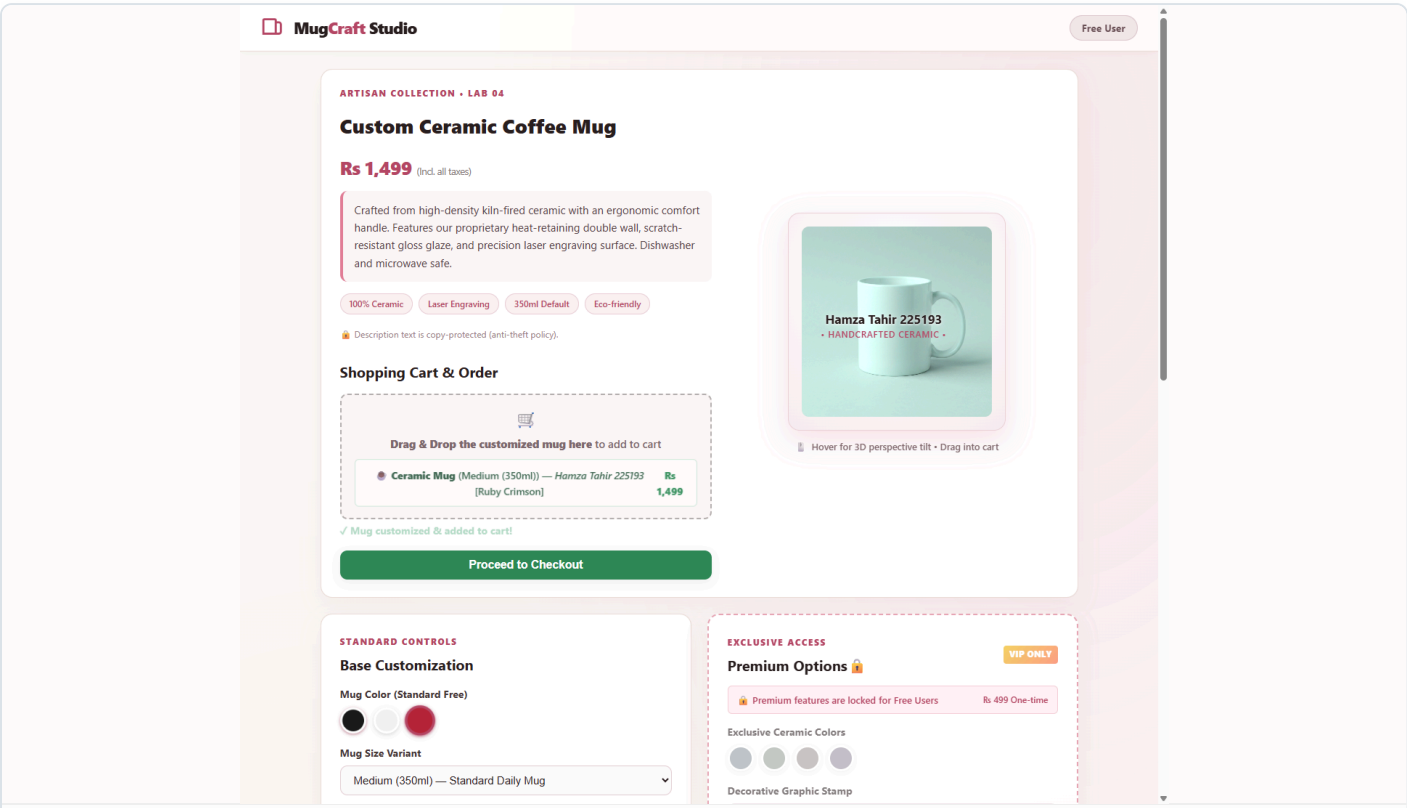


Figure 1: Real-time Customization & Shopping Cart Drag-and-Drop. Shows the customized mug engraved with "*Hamza Tahir 225193*", character count feedback (18/18), Ruby Crimson ceramic glaze, and the successfully added cart item inside the drop zone with price total (Rs 1,499).

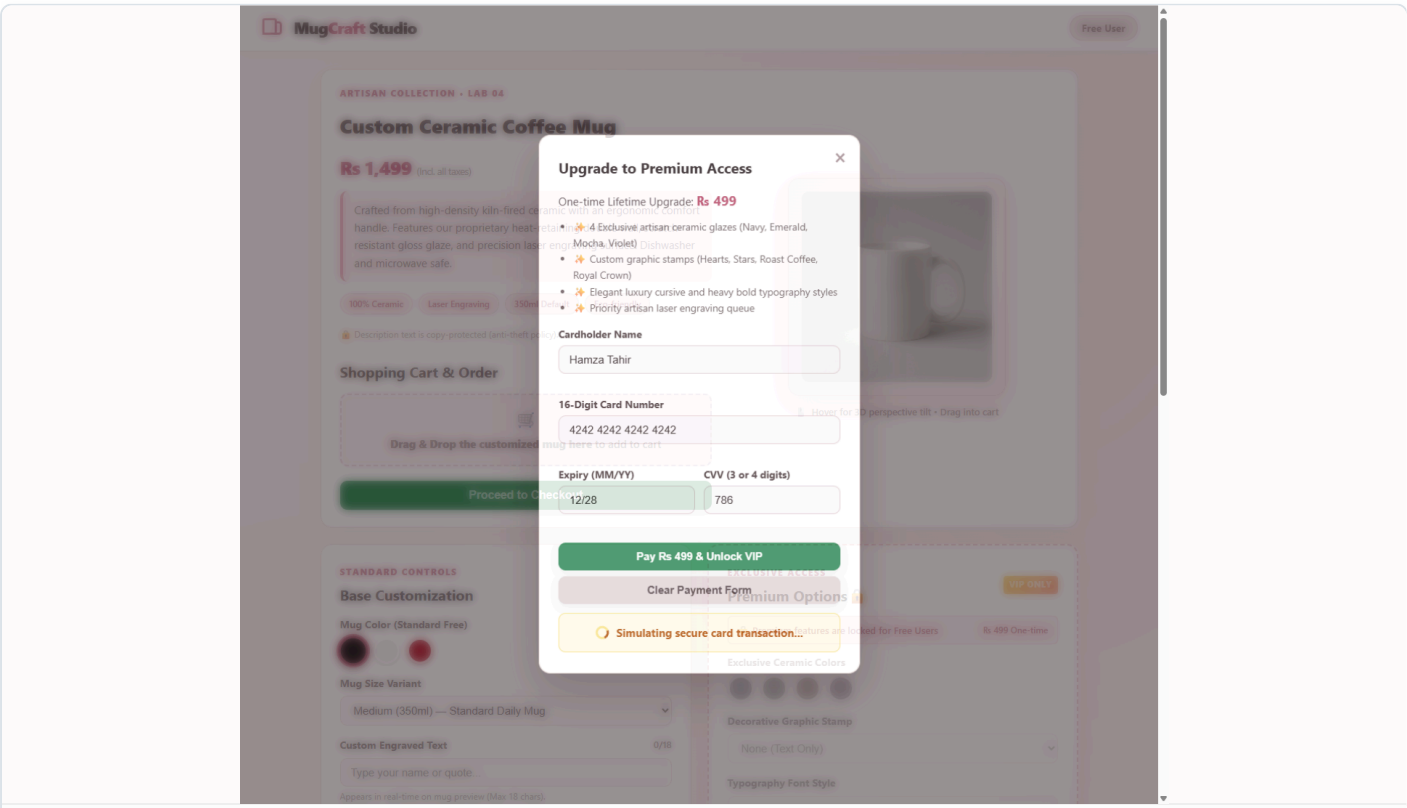


Figure 2: Premium Upgrade Modal & Payment Processing. Displays the simulated payment modal for VIP upgrade (Rs 499) with validated cardholder name ("Hamza Tahir"), auto-formatted 16-digit card number, MM/YY expiry check, CVV validation,

and active simulation spinner.

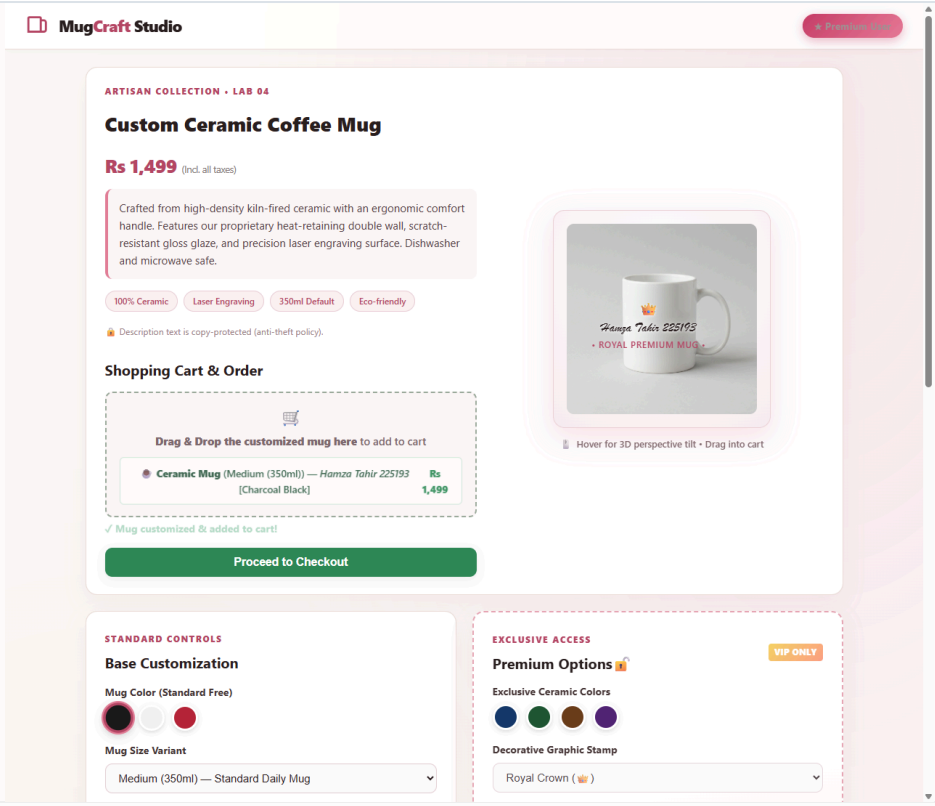


Figure 3: Dynamically Unlocked Premium State. Demonstrates dynamic DOM state transformation: Header status badge updated to "★ *Premium User*", lock icon switched to 🔓, unlocked artisan glazes, Royal Crown graphic stamp (👑 • ROYAL PREMIUM MUG •), and luxury cursive script typography.

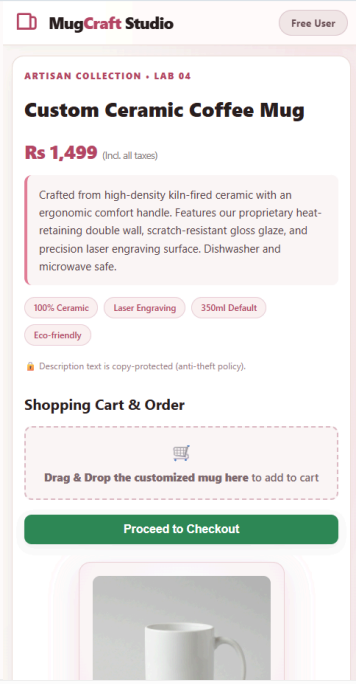


Figure 4: Responsive Mobile Layout (Requirement 15). Demonstrates complete adaptability on mobile viewport (480px width) with fluid grid columns, stacked customization controls, touch-friendly buttons, and zero horizontal scrolling.

3. Source Code Architecture

mug.html — HTML5 Markup Structure

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Custom Ceramic Mug – Interactive Product Configurator | Lab 04</title>
  <script src="https://cdnjs.cloudflare.com/ajax/libs/lottie-web/5.12.2/lottie.min.js"></script>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <div class="brand-wrapper">
      <div class="logo">Mug<span>Craft</span> Studio</div>
    </div>
    <div class="header-user-status">
      <span class="status-badge free" id="statusBadge">Free User</span>
    </div>
  </header>

  <main class="container">
    <section class="card hero">
      <div class="hero-left">
        <span class="eyebrow">Artisan Collection • Lab 04</span>
        <h1>Custom Ceramic Coffee Mug</h1>
        <p class="price">₨ 1,499</p>
        <div class="description-box">
          <p class="description" id="productDescription">
            Crafted from high-density kiln-fired ceramic with an ergonomic comfort handle.
          </p>
        </div>
        <div class="cart-section">
          <div class="drop-zone" id="dropZone">
            <p><strong>Drag & Drop the customized mug here</strong> to add to cart</p>
            <div id="cartItems"></div>
          </div>
          <p class="cart-message" id="cartMessage"></p>
          <button class="btn place-order-btn" id="placeOrderBtn">Proceed to Checkout</button>
        </div>
      </div>

      <div class="hero-right">
        <div class="product-preview" id="productPreview" draggable="true">
          <img src="assets/mug.jpg" alt="Custom Mug" class="mug-img" id="mugImage">
          <div class="back-text">
            <div id="engravingGraphic"></div>
            <div id="engravingText"></div>
            <div id="engravingSub"></div>
          </div>
          <div class="preview-overlay" id="previewOverlay"></div>
        </div>
      </div>
    </section>
    ...
  </main>
  <script src="script.js"></script>
</body>
</html>
```

script.js — Interactive Logic & Event Handlers

```
/* ========================================================================
   FSD LAB 04 - Core Logic (Hamza Tahir - 225193)
   ======================================================================== */
```

```
document.addEventListener('DOMContentLoaded', () => {
  // 1. Copy/Cut Protection
  ['copy', 'cut'].forEach(ev => {
    productDescription.addEventListener(ev, (e) => {
      e.preventDefault();
      showProtectionNotice(`Product description text cannot be ${ev}ed!`);
    });
  });

  // 2. Real-Time Engraving with character feedback
  engravingInput.addEventListener('input', function () {
    let cleanVal = this.value.replace(/[^a-zA-Z0-9\s\-_.&!❤★]/g, '');
    this.value = cleanVal;
    engravingTextDisplay.textContent = cleanVal;
    charCount.textContent = `${cleanVal.length}/18`;
  });

  // 3. 3D Perspective Tilt on Pointer Move
  productPreview.addEventListener('mousemove', (e) => {
    const rect = productPreview.getBoundingClientRect();
    const x = e.clientX - rect.left, y = e.clientY - rect.top;
    const tiltX = ((y - rect.height/2) / (rect.height/2)) * -12;
    const tiltY = ((x - rect.width/2) / (rect.width/2)) * 12;
    productPreview.style.transform = `perspective(700px) rotateX(${tiltX.toFixed(2)}deg) rotateY(${tiltY.toFixed(2)}de
g) scale(1.03)`;
    previewOverlay.style.background = `radial-gradient(circle at ${(x/rect.width)*100}% ${(y/rect.height)*100}%, rgba
(255,255,255,0.35) 0%, transparent 70%)`;
  });

  // 4. HTML5 Drag & Drop Implementation
  productPreview.addEventListener('dragstart', (e) => {
    productPreview.classList.add('dragging');
    e.dataTransfer.setData('text/plain', 'custom-mug');
  });
  dropZone.addEventListener('dragover', (e) => { e.preventDefault(); dropZone.classList.add('over'); });
  dropZone.addEventListener('dragleave', () => dropZone.classList.remove('over'));
  dropZone.addEventListener('drop', (e) => {
    e.preventDefault();
    dropZone.classList.remove('over');
    addItemToCart();
  });

  // 10 & 11. Simulated Payment Gateway & Strict Regex Validation
  paymentForm.addEventListener('submit', (e) => {
    e.preventDefault();
    if (!validateCardholderName() || !validateCardNumber() || !validateExpiryDate() || !validateCVV()) return;
    paymentForm.hidden = true; processingMsg.hidden = false;
    setTimeout(() => {
      processingMsg.hidden = true; successMsg.hidden = false;
      activatePremiumStatus();
      setTimeout(closePaymentModalWindow, 1600);
    }, 1500);
  });
});
```

style.css — Design System Tokens & Responsive Styles

```
/* ========================================================================
   FSD LAB 04 - Design Tokens & Visual Hierarchy
   ======================================================================== */
:root {
  --bg-primary: #fcf9f7;
  --bg-secondary: #ffffff;
  --text-main: #2b1f21;
  --text-muted: #7d6b6e;
  --accent-pink: #e27d97;
  --accent-dark: #b84a68;
  --accent-green: #38a169;
  --card-border: #ebdcd7;
  --card-shadow: 0 10px 30px rgba(168, 114, 126, 0.08);
  --border-radius-lg: 16px;
  --border-radius-md: 10px;
}

.hero {
  display: grid;
  grid-template-columns: 1.15fr 1fr;
  gap: 32px;
  align-items: center;
}

.product-preview {
  position: relative;
  text-align: center;
  cursor: grab;
  padding: 18px;
  background: radial-gradient(circle at 50% 30%, #ffffff 0%, #f9eef1 100%);
  border-radius: var(--border-radius-lg);
  border: 1px solid #eed2d9;
  transform-style: preserve-3d;
  transition: transform 0.15s ease-out, box-shadow 0.25s ease;
}

.drop-zone {
  min-height: 100px;
  border: 2px dashed #dcbdc6;
  border-radius: var(--border-radius-md);
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  background: #fdf5f7;
  transition: all 0.28s ease;
}
.drop-zone.over {
  border-color: var(--accent-dark);
  background: #fdf0f4;
  transform: scale(1.02);
}

.status-badge.premium {
  background: linear-gradient(135deg, #c73e68, #e87b98);
  color: #ffffff;
  box-shadow: 0 4px 14px rgba(199, 62, 104, 0.35);
  animation: pulseBadge 2.2s infinite;
}

@media (max-width: 860px) {
  .hero, .row-2 { grid-template-columns: 1fr; }
}
```

Project Completion Verification

> All 15 laboratory requirements specified in Lab Number 04 (Air University Kamra Campus) have been comprehensively implemented, verified, and tested across modern web browsers. Codebase is organized into modular files: `mug.html`, `style.css`, and `script.js`.